Rubén Coll Sánchez

Junior Researcher & Engineer – Software Performance & Computer Systems

Education

Sorbonne University VI - UPMC — **Paris, France**
MSc in Computer Science – Software Science & Technology — *Sep 2025 - Present*

University of Murcia — **Murcia, Spain**
BSc in Mathematics — *Sep 2019 - Jun 2024*

University of Murcia — **Murcia, Spain**
BSc in Computer Engineering — *Sep 2019 - Jun 2024*

Research Interests

My research interests intersect the areas of programming languages, computer architecture and high performance computing. I study how to characterize the performance and resource consumption of a system through formal methods and theoretical approaches.

Experience

Huawei Technologies France — **Boulogne-Billancourt, France**
HPC Research Engineer — *Jun 2025 - Present*
Research on performance models and semantics of tensor computations for distributed heterogeneous architectures.

- Co-led a research project on the theoretical design, implementation and validation of a lightweight collaborative framework for symbolic performance analysis, with direct application to distributed AI/ML computations.
- Implemented a novel algorithm for MPI-style collective communication on HPC systems with hierarchical topologies. Validated it on a cluster (384 accelerators), achieving sublinear speed-up over the existing best.

Huawei Technologies France — **Boulogne-Billancourt, France**
ML Systems Engineering Intern — *Sep 2024 - Jun 2025*
Performance analysis and optimization of distributed AI/ML training workloads on cluster architectures.

- Implemented a memory offloading scheduler that increased the effective memory budget by up to 33% on LLM training tasks with less than 2% time overhead verified tradeoff.
- Audited the internal deep-learning compiler framework, optimized the scheduling algorithms for recomputation of intermediate values and raised throughput by 6% on average on LLM training tasks.

University of Murcia — **Murcia, Spain**
AI Engineering Intern — *Nov 2023 - Jun 2024*
Design, training and deployment of a classifier multi-modal LLM for proteomic data analysis for drug discovery.

NaviLens — **Murcia, Spain**
Software Development Intern — *Jan 2023 - Sep 2023*
Development of a full-stack application for real-time transit data management and client-oriented integration tools.

Publications

Rubén Coll Sánchez, Thibaut Tachon, Pierre Leca, Teng Su and Chong Li: AMAI: Algebraic Modelling of Distributed AI Computations for Symbolic Performance Analysis. *Accepted at the 17th International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems - SC '26 Workshops*

Rubén Coll Sánchez, Hongxing Wang, Hector Buffière, Pierre Leca, Teng Su and Chong Li: Accelerating DNN Communications by Hierarchical Resharding. *Under review at the International Journal of Parallel Programming.*

Awards

Engineering Student of the Year 2024 — *Awarded by University of Murcia*

Academic Excellence Award - Computer Engineering — *Awarded by University of Murcia*

C2 – C2 – B2 – B1

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